# feb. 19

- **Warm-up**: I have a cyclic group of order 15 (say $G=\langle a\rangle$). Classify all the subgroups (diagram), include generators

Divisors of 15:
1, 3, 5, 15

$$\begin{tikzcd}[column sep=small]
 & G=\langle a^1\rangle \text{ order } 15 \arrow[dl, dash] \arrow[dr, dash] & \\
\langle a^3\rangle \text{ order } \frac{15}{3}=5 \arrow[dr, dash] \arrow[rr, bend right] & & \langle a^5\rangle \text{ order } 3=\frac{15}{5} \arrow[dl, dash] \arrow[ll, bend left] \\
 & \langle a^{15}\rangle=\langle e\rangle=\{e\} \text{ order } 1 &
\end{tikzcd}$$

Other generators:

$\langle a^5\rangle=\langle a^{10}\rangle$ $\gcd(k,15)=5$

$\langle a^3\rangle=\langle a^6\rangle=\langle a^9\rangle=\langle a^{12}\rangle$ $\gcd(k,15)=3$

$\langle a\rangle=\langle a^2\rangle=\langle a^4\rangle=\langle a^7\rangle=\langle a^8\rangle=\langle a^{11}\rangle=\langle a^{13}\rangle=\langle a^{14}\rangle$ all $\gcd(k,15)=1$

- Klein 4-Group: $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$
  - $(\bar{a},\bar{b})+(\bar{c},\bar{d})=(\bar{a}+\bar{c},\bar{b}+\bar{d})$
  - All subgroups cyclic, group is not (relevant to HW 3 Q.3)
  - Elements have order 1 or 2
  - Same structure as $\mathbb{Z}/8\mathbb{Z}^{\times}$

## Ch. 5: Permutation Groups

- Bijective functions (1-1 and onto) from a set to itself form a group

- **Def** A permutation of a set $X$ is a bijective function from $X$ to $X$
  - A permutation group of a set $X$ is a set of permutations of $X$ that forms a group under function composition

Ex. Suppose $X=\{1\}$. What are the functions from $X\to X$?

$\{1\} \to \{1\}$

$id: X\to X$ given by $id(1)=1$

The permutation group is simply $\{id\}$ ← $S_1$

Ex. Suppose $X=\{1,2\}$.

The set of all permutations of $\{1,2\}$ is $\{id,\sigma\}$ ← $S_2$

$1 \mapsto 1$, $2 \mapsto 2$: $id$

$1 \mapsto 2$, $2 \mapsto 1$: $\sigma$ given by $\sigma(1)=2$, $\sigma(2)=1$

- **Def** The group of all permutations of a set with $n$ elements (we usually label $1,\dots,n$) is called the symmetric group of degree $n$. This group is denoted $S_n$.

- Why is this a group? Example/Justification with $S_4$

If $\alpha, \beta : \{1,2,3,4\} \xrightarrow{\text{bijective}} \{1,2,3,4\}$, then Closure

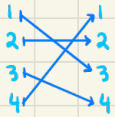
$\alpha \circ \beta : \{1,2,3,4\} \xrightarrow{\text{bijective}} \{1,2,3,4\}$

yep. proof by because-I-said-so ^v^

★ function composition is associative (by decree)

$id : \{1,2,3,4\} \xrightarrow{\text{bijective}} \{1,2,3,4\}$ identity

$\alpha$:

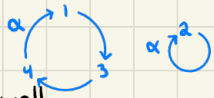



$\alpha^{-1}$:

Inverses

ex $\alpha^{-1} \circ \alpha(3) = \alpha^{-1}(4) = 3$

- What is a nice way to write permutations?
  - $\alpha : \begin{matrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{matrix}$, $\alpha \circ \alpha = \alpha^2 = \begin{matrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{matrix}$ bad notation, but some people do
  - As in example above?
  - Notice: the permutations cycle




  - <u>Cycle notation</u> illustrates order well

↪ $\alpha = (134)(2) = (341)(2)$ etc, doesn't matter where you start each cycle

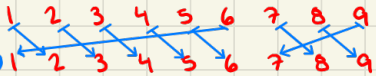
$\alpha \circ \alpha = \alpha^2 = (143)(2)$ notice it does matter what order they are in, just not the starting point

Single cycles sometimes omitted, just means perm. acts as identity there

- <u>Cycle Notation</u>

<u>Ex</u> $\beta \in S_9$

$\beta = (123456)(798)$




6-cycle or six-cycle

Disjoint cycles (don't share elements)

find order of $\beta$ with lcm of cycle sizes